How Bloom Filters Work

To fully comprehend how Bloom filters work, one must delve into their components, the process of adding elements, querying elements, and the occurrence of false positives and false negatives.

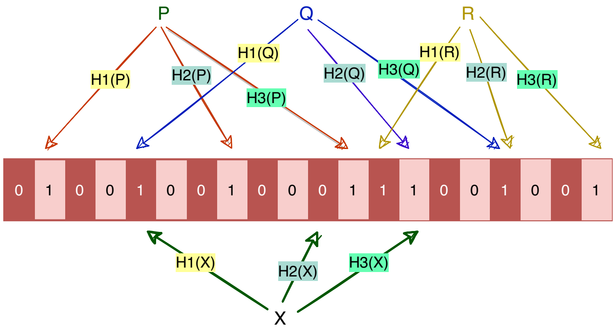
A. Components: Bit Array and Hash Functions

A Bloom filter consists of two primary components: a bit array and a collection of hash functions. The bit array is a fixed-size sequence of bits (0 or 1) initialized to all zeroes. The number of hash functions, usually denoted as 'k', determines how many positions in the bit array an element maps to. The hash functions should ideally be independent and uniformly distributed to minimize the probability of false positives.

B. Adding Elements to the Filter

To add an element to the Bloom filter, the element is passed through each of the 'k' hash functions, generating 'k' different hash values. Each hash value corresponds to a position in the bit array. The bits at these positions are then set to 1. This process is repeated for all elements that need to be added to the filter.

For example, following Bloom filter consists of 20 bits and uses three (k=3) hash functions `H1`, `H2`, and `H3`.



Bloom Filter Using 3 Hash Functions

C. Querying Elements in the Filter

To check if an element is a member of the set, the same 'k' hash functions are applied to the element, generating 'k' positions in the bit array. If any of the bits at these positions are 0, the element is not in the set (no false negatives). However, if all the bits at these positions are 1, the element is considered to be in the set, but there's a possibility of a false positive (the bits might have been set to 1 by other elements).

D. False Positives and False Negatives

Bloom filters guarantee no false negatives, but they may produce false positives. The probability of a false positive depends on the size of the bit array (m), the number of hash functions (k), and the number of elements inserted into the filter (n). As the filter becomes more populated, the probability of false positives increases.

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